Congratulations for seeking Govt. Sponsored Project

Name:	Dr. Anindita Bera
Department:	Mathematics
Project Title:	**Construction of entanglement witnesses exploiting positive maps and their optimization procedures.**
Funding Agency:	National Board for Higher Mathematics (NBHM), Department of Atomic Energy (DAE), Govt. of India
Sanction Amount:	Rs. 1959200
Tenure:	3 Years

Abstract:

Entanglement is one of the central concepts of quantum information theory. Beyond its fundamental significance, it plays a crucial operational role in a wide range of quantum information–processing tasks, including quantum communication, computation, and metrology. Many advantages of quantum technologies over their classical counterparts can ultimately be traced back to the presence of entanglement.

From a mathematical point of view, entanglement is defined through a simple structural criterion: a quantum state is entangled if it cannot be written as a convex combination of product states. While this definition is conceptually clear, its practical implementation becomes extremely challenging as the dimension of the underlying Hilbert space increases. In high-dimensional or multipartite systems, a full characterization of the state typically requires an exponentially large number of parameters, rendering standard entanglement detection methods infeasible. Moreover, commonly used criteria such as the positive partial transpose condition are no longer sufficient to detect all entangled states in higher dimensions, leading to the phenomenon of bound entanglement.

The goal of this project is to address these challenges by designing entanglement witnesses tailored for high-dimensional quantum systems. In particular, the project will exploit the theory of positive but not completely positive maps, which provides a powerful and systematic framework for entanglement detection beyond low-dimensional settings. We further aim to develop experimentally accessible and computationally efficient tools for certifying entanglement in complex quantum systems. These results are expected to contribute both to the theoretical understanding of high-dimensional entanglement and to its practical verification in realistic quantum information platforms.